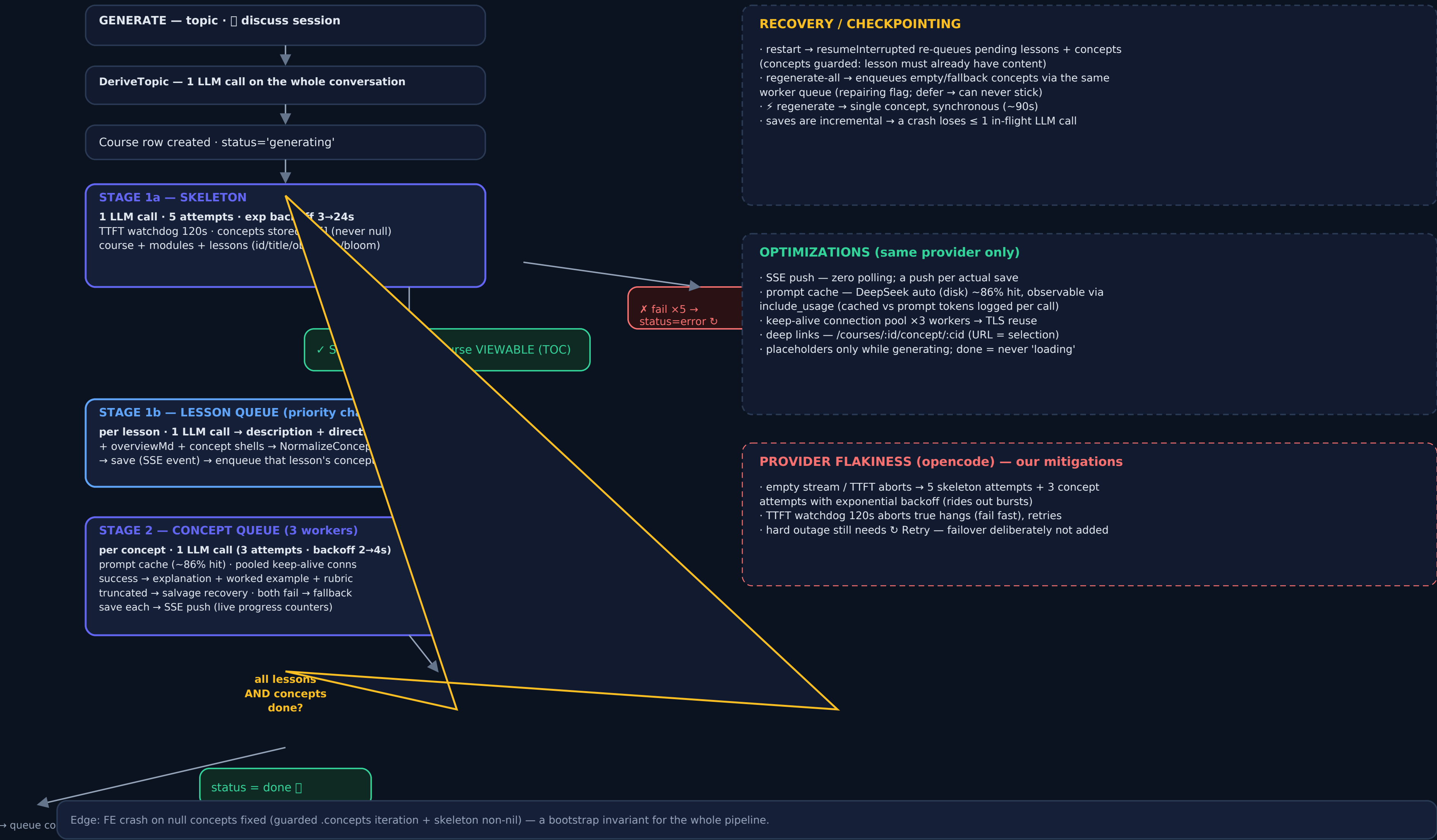


Trace — Course Generation Pipeline (2026)

async 3-stage generation · SSE push · prompt cache · restart-safe



GENERATE — topic · discuss session

DeriveTopic — 1 LLM call on the whole conversation

Course row created · status='generating'

STAGE 1a — SKELETON

1 LLM call · 5 attempts · exp backoff 3→24s
TTFT watchdog 120s · concepts stored in DB (never null)
course + modules + lessons (id/title/output/bloom)

✓ Skeleton course VIEWABLE (TOC)

STAGE 1b — LESSON QUEUE (priority change)

per lesson · 1 LLM call → description + direct
+ overviewMd + concept shells → NormalizeConcept
→ save (SSE event) → enqueue that lesson's concept

STAGE 2 — CONCEPT QUEUE (3 workers)

per concept · 1 LLM call (3 attempts · backoff 2→4s)
prompt cache (~86% hit) · pooled keep-alive conns
success → explanation + worked example + rubric
truncated → salvage recovery · both fail → fallback
save each → SSE push (live progress counters)

all lessons
AND concepts
done?

status = done

RECOVERY / CHECKPOINTING

- restart → resume interrupted re-queues pending lessons + concepts (concepts guarded: lesson must already have content)
- regenerate-all → enqueues empty/fallback concepts via the same worker queue (repairing flag; defer → can never stick)
- regenerate → single concept, synchronous (~90s)
- saves are incremental → a crash loses ≤ 1 in-flight LLM call

OPTIMIZATIONS (same provider only)

- SSE push — zero polling; a push per actual save
- prompt cache — DeepSeek auto (disk) ~86% hit, observable via include_usage (cached vs prompt tokens logged per call)
- keep-alive connection pool x3 workers → TLS reuse
- deep links — /courses/:id/concept/:cid (URL = selection)
- placeholders only while generating; done = never 'loading'

PROVIDER FLAKINESS (opencode) — our mitigations

- empty stream / TTFT aborts → 5 skeleton attempts + 3 concept attempts with exponential backoff (rides out bursts)
- TTFT watchdog 120s aborts true hangs (fail fast), retries
- hard outage still needs Retry — failover deliberately not added

Edge: FE crash on null concepts fixed (guarded .concepts iteration + skeleton non-nil) — a bootstrap invariant for the whole pipeline.